

DR ESLAM AHMED
eliteigcse.com

Graphs of Functions

Know the shapes, mark the crossings, and count intersections with a discriminant. Sketch first — the picture tells you the algebra.

THE TOPIC ON ONE MAP

Graphs of Functions

01 — Cubic sketches

$$(2x + 5)(x - 3)^2 \quad f(x) \leq 0$$

02 — Reciprocal graphs

$$y = \frac{k}{x} \quad y = \frac{k}{x^2} \quad \text{asymptote}$$

03 — Intersections and roots

coordinates of P, Q number of roots

04 — Tangent / cross / miss

$$b^2 - 4ac \quad \text{at least one point}$$

01 Cubic Sketches from Factorised Form

Roots from the brackets, y-intercept from $x = 0$, shape from the sign of x^3 .
A squared bracket **touches**.

WHEN YOU SEE → YOU DO

“sketch $y = (2x + 5)(x - 3)^2$ ”

→ cuts at $-\frac{5}{2}$, touches at 3, y-int $f(0)$

“deduce the values of x for which $f(x) \leq 0$ ”

→ read below the axis from the sketch

“expand to $ax^3 + bx^2 + cx + d$ ” then
“gradient at P ”

→ differentiate the expansion

“translated two units in the positive x direction”

→ $g(x) = f(x - 2)$

WORKED MODEL • OCTOBER 2019

$$f(x) = (2x + 5)(x - 3)^2$$

1 cuts at $x = -\frac{5}{2}$, touches at $x = 3$

brackets

2 $f(x) = 2x^3 - 7x^2 - 12x + 45$, $P(0, 45)$

expand, $x = 0$

ANSWER $f(x) \leq 0 \iff x \leq -\frac{5}{2} \text{ or } x = 3$

IDEA 01

Practise: Cubic Sketches

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Squared bracket touches**P1 · TOPIC 6** **3 marks**

Sketch $y = (x + 1)(x - 2)^2$, showing where it meets the axes.

SELECTED PRACTICE**Q2** **PRACTICE** Practice · Easy**P1 · TOPIC 6** **5 marks**

6 (a) Express $2x^3 + 2x^2 - 12x$ in the form $ax(x + b)(x + c)$, where a , b and c are integers to be found.

(2 marks)

(b) Hence sketch the graph of $y = 2x^3 + 2x^2 - 12x$ labelling any points where the graph intersects the coordinate axes.

(3 marks)

Q3**PRACTICE**

Practice · Medium

P1 · TOPIC 6**3 marks**

- 1 Sketch the graph of $y = (x + 3)^3$ labelling any points where the graph intersects the coordinate axes.

(3 marks)**Q4****PRACTICE**

Practice · Medium

P1 · TOPIC 6**3 marks**

- 5 Sketch the graph of $y = 2x^2(x + 3)$ labelling any points where the graph intersects the coordinate axes.

(3 marks)

PAST PAPERS

Q5 PAST PAPER Pearson IAL WMA11 · October 2019 · Q10

P1 · TOPIC 6 10 marks

10.

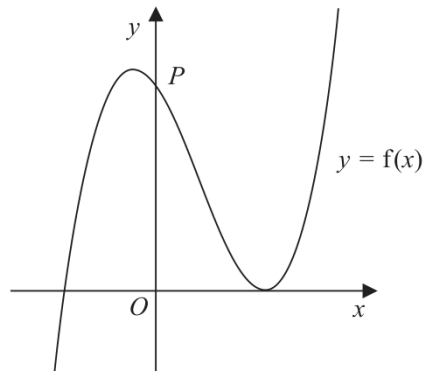


Figure 6

Figure 6 shows a sketch of part of the curve with equation $y = f(x)$, where

$$f(x) = (2x + 5)(x - 3)^2$$

- (a) Deduce the values of x for which $f(x) \leq 0$ (2)

The curve crosses the y -axis at the point P , as shown.

- (b) Expand $f(x)$ to the form

$$ax^3 + bx^2 + cx + d$$

where a , b , c and d are integers to be found.

(3)

- (c) Hence, or otherwise, find

- (i) the coordinates of P ,
 (ii) the gradient of the curve at P . (2)

The curve with equation $y = f(x)$ is translated two units in the positive x direction to a curve with equation $y = g(x)$.

- (d) (i) Find $g(x)$, giving your answer in a simplified factorised form.
 (ii) Hence state the y intercept of the curve with equation $y = g(x)$. (3)

(Total 10 marks)

Q6 PAST PAPER Pearson IAL WMA11 · January 2020 · Q10

P1 · TOPIC 6 8 marks

10. The curve C_1 has equation $y = f(x)$, where

$$f(x) = (4x - 3)(x - 5)^2$$

- (a) Sketch C_1 showing the coordinates of any point where the curve touches or crosses the coordinate axes.

(3)

(b) Hence or otherwise

- (i) find the values of x for which $f\left(\frac{1}{4}x\right) = 0$

- (ii) find the value of the constant p such that the curve with equation $y = f(x) + p$ passes through the origin.

(2)

A second curve C_2 has equation $y = g(x)$, where $g(x) = f(x + 1)$

- (c) (i) Find, in simplest form, $g(x)$. You may leave your answer in a factorised form.

- (ii) Hence, or otherwise, find the y intercept of curve C_2

(3)

(Total 8 marks)

Q7**PAST PAPER**

Pearson IAL WMA11 · January 2021 · Q8

P1 · TOPIC 6 9 marks

8.

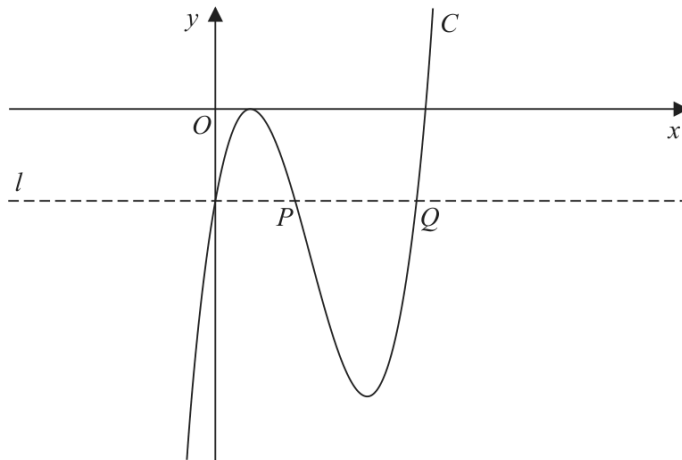
**Figure 4**

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$, where

$$f(x) = (3x - 2)^2 (x - 4)$$

- (a) Deduce the values of x for which $f(x) > 0$ (1)

- (b) Expand $f(x)$ to the form

$$ax^3 + bx^2 + cx + d$$

where a , b , c and d are integers to be found.

(3)

The line l , also shown in Figure 4, passes through the y intercept of C and is parallel to the x -axis.

The line l cuts C again at points P and Q , also shown in Figure 4.

- (c) Using algebra and showing your working, find the length of line PQ . Write your answer in the form $k\sqrt{3}$, where k is a constant to be found.

(Solutions relying entirely on calculator technology are not acceptable.)

(5)**(Total 9 marks)**

Q8 PAST PAPER Pearson IAL WMA11 · October 2021 · Q6

P1 · TOPIC 6 10 marks

6. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A curve C has equation $y = f(x)$ where

$$f(x) = 2(x + 1)(x - 3)^2$$

(a) Sketch a graph of C .

Show on your graph the coordinates of the points where C cuts or meets the coordinate axes.

(3)

(b) Write $f(x)$ in the form $ax^3 + bx^2 + cx + d$, where a, b, c and d are constants to be found.

(3)

(c) Hence, find the equation of the tangent to C at the point where $x = \frac{1}{3}$

(4)

(Total 10 marks)

Q9**PAST PAPER**

Pearson IAL WMA11 · June 2025 · Q7

P1 · TOPIC 6**10 marks**

7.

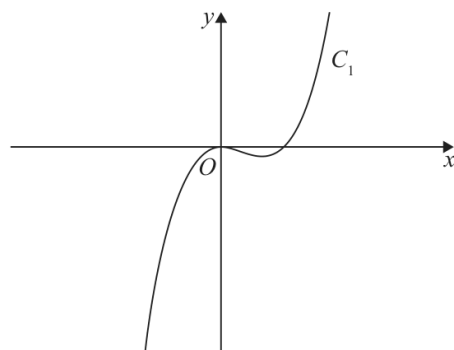
**Figure 3**

Figure 3 shows a sketch of part of the curve C_1

Given that C_1

- has equation $y = f(x)$ where $f(x)$ is a cubic function
- touches the x -axis at the origin and cuts the x -axis at $x = 4$
- passes through the point $(10, 120)$

(a) find $f(x)$

(3)

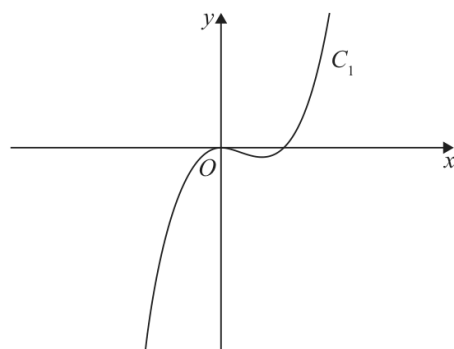
The curve C_2 has equation $y = 1.2x(8 - x)$

On the following page there is a copy of Figure 3 called Diagram 1.

(b) On Diagram 1 sketch a graph of the curve C_2

(2)

(c) Use algebra to find the coordinates of the points where C_1 and C_2 intersect.
Show each stage of your working.

(5)**Diagram 1****(Total for Question 7 is 10 marks)**

02 Reciprocal Graphs $y = \frac{k}{x}$, $y = \frac{k}{x^2}$

Two branches. Asymptotes: where the bottom is zero, and the number added on.

WHEN YOU SEE → YOU DO

$$y = \frac{4}{x - k}$$

→ vertical asymptote $x = k$; y-int $-\frac{4}{k}$

$$y = \frac{4}{x} + k$$

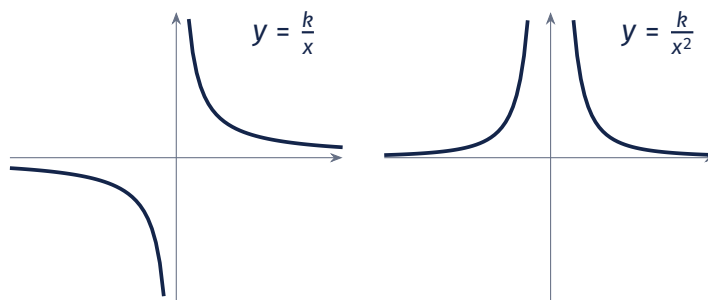
→ horizontal asymptote $y = k$; x-int $-\frac{4}{k}$

$$y = \frac{k}{x^2}$$

→ both branches above the axis

“state the equation of the asymptote”

→ write $x = \dots$ or $y = \dots$



IDEA 02

Practise: Reciprocal Graphs

My questions first, then selected practice, then real past papers.

Q10 DR ESLAM Asymptotes

P1 · TOPIC 6 2 marks

State the asymptotes of (a) $y = \frac{3}{x-2}$ (b) $y = \frac{1}{x} + 4$

SELECTED PRACTICE**Q11** PRACTICE Practice · Easy

P1 · TOPIC 6 3 marks

- 3 Sketch the graph of $y = \frac{1}{x}$, labelling any points where the graph intersects the coordinate axes and stating the equations of any asymptotes.

(3 marks)

Q12**PRACTICE**

Practice · Medium

P1 · TOPIC 6 2 marks

- 2 Sketch the graph of $y = \frac{1}{x^2}$ and write down the equations of any asymptotes.

(2 marks)**Q13****PRACTICE**

Practice · Medium

P1 · TOPIC 6 3 marks

- 4 y is inversely proportional to x . When $x = 3$, $y = 12$. Find the constant of proportionality and sketch the graph of y against x .

(3 marks)

PAST PAPERS

Q14 PAST PAPER Pearson IAL WMA11 · January 2021 · Q6

P1 · TOPIC 6 10 marks

6. (a) Sketch the curve with equation

$$y = -\frac{k}{x} \quad k > 0 \quad x \neq 0 \quad (2)$$

- (b) On a separate diagram, sketch the curve with equation

$$y = -\frac{k}{x} + k \quad k > 0 \quad x \neq 0$$

stating the coordinates of the point of intersection with the x -axis and, in terms of k , the equation of the horizontal asymptote.

(3)

- (c) Find the range of possible values of
- k
- for which the curve with equation

$$y = -\frac{k}{x} + k \quad k > 0 \quad x \neq 0$$

does not touch or intersect the line with equation $y = 3x + 4$

(5)

(Total 10 marks)

Q15**PAST PAPER**

Pearson IAL WMA11 · October 2022 · Q6

P1 · TOPIC 6 6 marks

6. (a) Given that k is a positive constant such that $0 < k < 4$ sketch, on **separate axes**, the graphs of

(i) $y = (2x - k)(x + 4)^2$

(ii) $y = \frac{k}{x^2}$

showing the coordinates of any points where the graphs cross or meet the coordinate axes, leaving coordinates in terms of k , where appropriate.

(5)

- (b) State, with a reason, the number of roots of the equation

$$(2x - k)(x + 4)^2 = \frac{k}{x^2}$$

(1)**(Total for Question 6 is 6 marks)**

Q16**PAST PAPER**

Pearson IAL WMA11 • October 2023 • Q4

P1 • TOPIC 6 7 marks

4.

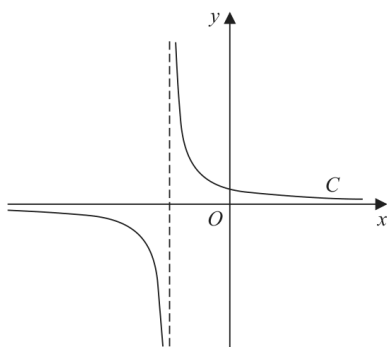
**Figure 1**

Figure 1 shows a sketch of part of the curve C with equation $y = \frac{1}{x+2}$

(a) State the equation of the asymptote of C that is parallel to the y -axis. (1)

(b) Factorise fully $x^3 + 4x^2 + 4x$ (2)

A copy of Figure 1, labelled Diagram 1, is shown on the next page.

(c) On Diagram 1, add a sketch of the curve with equation

$$y = x^3 + 4x^2 + 4x$$

On your sketch, state clearly the coordinates of each point where this curve cuts or meets the coordinate axes.

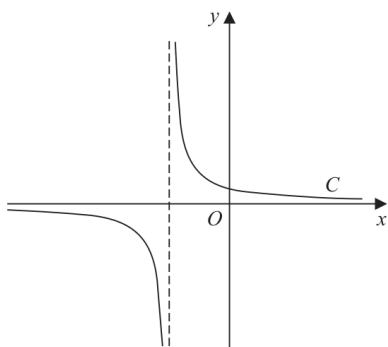
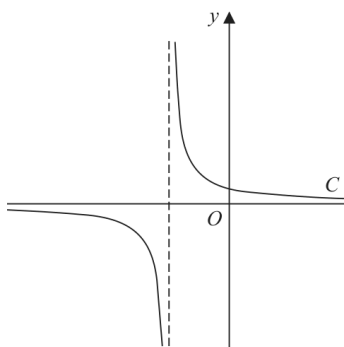
(3)

(d) Hence state the number of real solutions of the equation

$$(x+2)(x^3 + 4x^2 + 4x) = 1$$

giving a reason for your answer.

(1)

**Diagram 1****copy of Diagram 1**

Only use the copy of Diagram 1 if you need to redraw your answer to part (c).

(Total for Question 4 is 7 marks)

03 Intersections and How Many Roots

Where two graphs meet: set them equal. How many solutions: count the crossings on a sketch.

WHEN YOU SEE → YOU DO

“find the coordinates of P and Q ”

→ curve = line → solve → pair

“state the number of roots, with a reason”

→ sketch both sides, count crossings

“use inequalities to define R ”

→ one inequality per edge

“the line passes through the y -intercept, parallel to the x -axis”

→ $y = f(0)$

WORKED MODEL • JANUARY 2022

$$x + y = 6 \text{ and } y = 6x - 2x^2 + 1$$

1 $6 - x = 6x - 2x^2 + 1 \Rightarrow 2x^2 - 7x + 5 = 0$

equal

2 $x = 1, \frac{5}{2}$

solve

ANSWER $P(1, 5), Q\left(\frac{5}{2}, \frac{7}{2}\right)$

IDEA 03

Practise: Intersections

My questions first, then selected practice, then real past papers.

SELECTED PRACTICE**Q17****PRACTICE**

Practice · Easy

P1 · TOPIC 6 3 marks

- 8 By sketching the graphs of $y = x^3$ and $y = \frac{1}{x}$ on the same diagram show that there are two real solutions to the equation $x^3 = \frac{1}{x}$.

(3 marks)

Q18**PRACTICE**

Practice · Medium

P1 · TOPIC 6 4 marks

6 (a) On the same diagram, sketch the graphs of $y = x(x+2)(x-1)$ and $y = \frac{1}{x}$.

(3 marks)

(b) Use your graph to determine the number of solutions to the equation

$$x(x+2)(x-1) = \frac{1}{x}.$$

(1 mark)

PAST PAPERS

Q19 PAST PAPER Pearson IAL WMA11 · October 2019 · Q3

P1 · TOPIC 6 10 marks

3. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

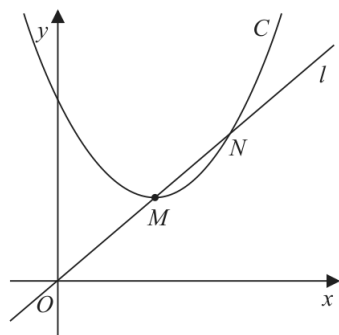


Figure 2

Figure 2 shows a sketch of the curve C with equation $y = x^2 - 5x + 13$

The point M is the minimum point of C .

The straight line l passes through the origin O and intersects C at the points M and N as shown.

Find, showing your working,

- (a) the coordinates of M , (3)
(b) the coordinates of N . (5)

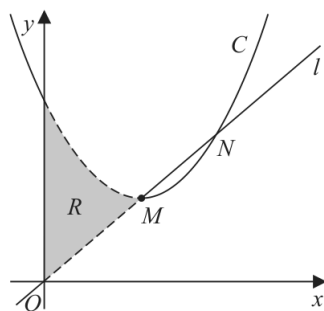


Figure 3

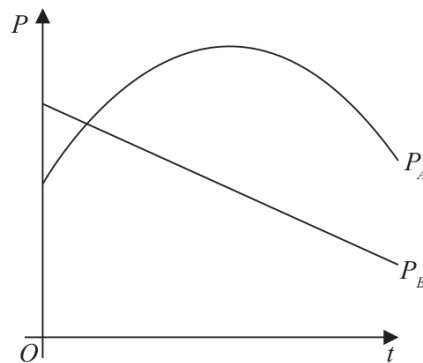
Figure 3 shows the curve C and the line l . The finite region R , shown shaded in Figure 3, is bounded by C , l and the y -axis.

(Total 10 marks)

Q20 PAST PAPER Pearson IAL WMA11 · June 2021 · Q5

P1 · TOPIC 6 8 marks

5.

**Figure 2**

The share value of two companies, company *A* and company *B*, has been monitored over a 15-year period.

The share value P_A of **company A**, in millions of pounds, is modelled by the equation

$$P_A = 53 - 0.4(t - 8)^2 \quad t \geq 0$$

where t is the number of years after monitoring began.

The share value P_B of **company B**, in millions of pounds, is modelled by the equation

$$P_B = -1.6t + 44.2 \quad t \geq 0$$

where t is the number of years after monitoring began.

Figure 2 shows a graph of both models.

Use the equations of one or both models to answer parts (a) to (d).

- (a) Find the difference between the share value of **company A** and the share value of **company B** at the point monitoring began. (2)
- (b) State the maximum share value of **company A** during the 15-year period. (1)
- (c) Find, using algebra and showing your working, the times during this 15-year period when the share value of **company A** was greater than the share value of **company B**. (4)
- (d) Explain why the model for **company A** should not be used to predict its share value when $t = 20$ (1)

(Total 8 marks)

4.

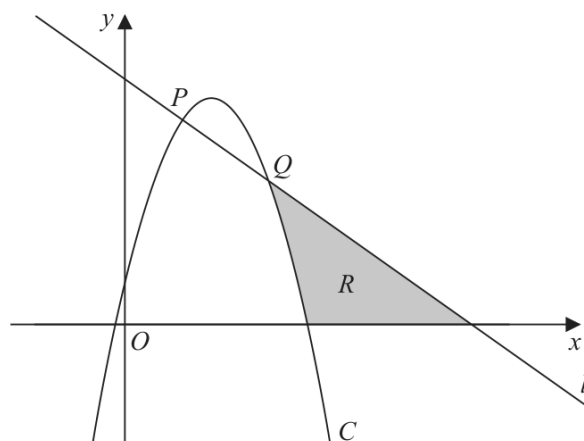


Figure 1

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Figure 1 shows a line l with equation $x + y = 6$ and a curve C with equation $y = 6x - 2x^2 + 1$

The line l intersects the curve C at the points P and Q as shown in Figure 1.

- (a) Find, using algebra, the coordinates of P and the coordinates of Q . (4)

The region R , shown shaded in Figure 1, is bounded by C , l and the x -axis.

- (b) Use inequalities to define the region R . (3)

(Total 7 marks)

04 Tangent / Cross / Miss: the Discriminant

Line meets a reciprocal or quadratic: multiply out, rearrange to a quadratic, then use $b^2 - 4ac$.

WHEN YOU SEE → YOU DO

"the line is a tangent to C "	→ $b^2 - 4ac = 0$
"does not touch or intersect"	→ $b^2 - 4ac < 0$
"intersects at two distinct points"	→ $b^2 - 4ac > 0$
"at least one point of intersection"	→ $b^2 - 4ac \geq 0$
" k is a positive constant"	→ keep only $k > 0$ in the final range
"meets at 4 distinct points" (x^4)	→ $u = x^2$: two positive roots in u

WORKED MODEL • OCTOBER 2019

$y = \frac{4}{x} + k$ and the tangent $y = 10 - 2x$.

1 $4 + kx = 10x - 2x^2 \Rightarrow 2x^2 + (k - 10)x + 4 = 0$

× x , rearrange

2 $(k - 10)^2 - 32 = 0$

tangent

ANSWER $k = 10 \pm 4\sqrt{2}$

IDEA 04

Practise: Tangent, Cross or Miss

My questions first, then real past papers.

Q22 DR ESLAM Tangent

P1 · TOPIC 6 3 marks

Find k so that the line $y = 3x - k$ is a tangent to $y = x^2$.

$k =$ _____

PAST PAPERS

Q23 PAST PAPER Pearson IAL WMA11 · October 2019 · Q6

P1 · TOPIC 6 8 marks

6. The curve C has equation $y = \frac{4}{x} + k$, where k is a positive constant.
- (a) Sketch a graph of C , stating the equation of the horizontal asymptote and the coordinates of the point of intersection with the x -axis. (3)

The line with equation $y = 10 - 2x$ is a tangent to C .

- (b) Find the possible values for k . (5)

(Total 8 marks)

Q24 **PAST PAPER** Pearson IAL WMA11 · January 2022 · Q10**P1 · TOPIC 6** **9 marks**

10. The curve C has equation

$$y = \frac{1}{x^2} - 9$$

(a) Sketch the graph of C .

On your sketch

- show the coordinates of any points of intersection with the coordinate axes
- state clearly the equations of any asymptotes

(4)

The curve D has equation $y = kx^2$ where k is a constant.

Given that C meets D at 4 distinct points,

(b) find the range of possible values for k .

(5)

(Total 9 marks)

Q25 **PAST PAPER** Pearson IAL WMA11 · January 2024 · Q7**P1 · TOPIC 6** **9 marks**

7. (a) Sketch the graph of the curve C with equation

$$y = \frac{4}{x - k}$$

where k is a positive constant.

Show on your sketch

- the coordinates of any points where C cuts the coordinate axes
- the equation of the vertical asymptote to C

(4)

Given that the straight line with equation $y = 9 - x$ does not cross or touch C

- (b) find the range of values of k .

(5)

(Total for Question 7 is 9 marks)

Q26 **PAST PAPER** Pearson IAL WMA11 · June 2024 · Q8**P1 · TOPIC 6** **7 marks**

8. The curve C_1 has equation

$$y = x(4 - x^2)$$

- (a) Sketch the graph of C_1 showing the coordinates of any points of intersection with the coordinate axes.

(3)

The curve C_2 has equation $y = \frac{A}{x}$ where A is a constant.

- (b) Show that the x coordinates of the points of intersection of C_1 and C_2 satisfy the equation

$$x^4 - 4x^2 + A = 0$$

(1)

- (c) Hence find the range of possible values of A for which C_1 meets C_2 at 4 distinct points.

(3)**(Total for Question 8 is 7 marks)**

Q27**PAST PAPER**

Pearson IAL WMA11 · October 2024 · Q6

P1 · TOPIC 6 9 marks

6. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Sketch the curve C with equation

$$y = \frac{1}{2-x} \quad x \neq 2$$

State on your sketch

- the equation of the vertical asymptote
- the coordinates of the intersection of C with the y -axis

(3)

The straight line l has equation $y = kx - 4$, where k is a constant.

Given that l cuts C at least once,

(b) (i) show that

$$k^2 - 5k + 4 \geq 0$$

(ii) find the range of possible values for k .

(6)

(Total for Question 6 is 9 marks)

Q28**PAST PAPER**

Pearson IAL WMA11 · January 2025 · Q7

P1 · TOPIC 6 8 marks

7. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation

$$y = \frac{2}{x} - k$$

where k is a **positive** constant.

- (a) Sketch the graph of C .

Show on your sketch

- the coordinates of any points of intersection of C with the coordinate axes
- the equation of the horizontal asymptote to C

stating each in terms of k .

(3)

The line l has equation $y = -kx - 6$

Given that l intersects C at 2 distinct points,

- (b) find the range of possible values of k .

(5)

(Total for Question 7 is 8 marks)

Q29 **PAST PAPER** Pearson IAL WMA11 · October 2025 · Q6**P1 · TOPIC 6** **9 marks**

6. (a) Sketch the graph of the curve C with equation

$$y = \frac{4k}{x - 2k}$$

where k is a positive constant.

On your sketch show

- the coordinates of any points where C cuts the coordinate axes
- the equation of the vertical asymptote to C

(4)

The straight line l has equation

$$y = 6 - 2x$$

Given that there is at least one point of intersection between l and C ,

(b) find the range of possible values of k .

(5)

(Total for Question 6 is 9 marks)

Q30 **PAST PAPER** Pearson IAL WMA11 · June 2026 · Q10**P1 · TOPIC 6** **11 marks**

10. The curve C has equation

$$y = \frac{3k}{x - 2k} \quad x \neq 2k$$

where k is a positive constant.

(a) Sketch a graph of C .

Show on your sketch the coordinates of the point where C crosses a coordinate axis.

(3)

The line l has equation $x + y = 12$

Given that l is a tangent to C ,

(b) show that

$$k^2 - 15k + 36 = 0$$

(4)

The line l meets C at a point P .

Given that P lies below the x -axis,

(c) find the coordinates of P .

(Solutions relying entirely on calculator technology are not acceptable.)

(4)

Question 10 continued

(Total for Question 10 is 11 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|----------------------|---|
| 01 | Cubics | Roots from brackets, squared bracket touches, y-int at $x = 0$. |
| 02 | Reciprocals | Asymptotes: bottom = 0, and the added number. |
| 03 | Intersections | Set equal $\rightarrow$ solve $\rightarrow$ pair. Count crossings for "how many". |
| 04 | Discriminant | Tangent = 0 • miss < 0 • two points > 0. Keep $k > 0$. |

BOTTOM LINE Sketch $\rightarrow$ set equal $\rightarrow$ quadratic $\rightarrow$ discriminant decides tangent, cross or miss.

Answers

Graphs of Functions • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • CUBIC SKETCHES

- Q1** Dr Eslam • Squared bracket touches cuts at $(-1, 0)$, touches at $(2, 0)$, y-int $(0, 4)$
- Q2** Practice • Easy (a) $2x(x + 3)(x - 2)$ (b) cuts at $-3, 0, 2$
- Q3** Practice • Medium touches/crosses at $(-3, 0)$ (point of inflection), y-int $(0, 27)$
- Q4** Practice • Medium touches at $(0, 0)$, cuts at $(-3, 0)$
- Q5** Pearson IAL WMA11 • October 2019 • Q10 (a) $x \leq -\frac{5}{2}$ or $x = 3$ (b) $2x^3 - 7x^2 - 12x + 45$ (c) $P(0, 45)$, gradient -12 (d) $g(x) = (2x+1)(x-5)^2$; y-int 25
- Q6** Pearson IAL WMA11 • January 2020 • Q10 (a) touches $(5, 0)$, cuts $(\frac{3}{4}, 0)$, $(0, -75)$ (b) $x = 3, 20$; $p = 75$ (c) $g(x) = (4x + 1)(x - 4)^2$; 16
- Q7** Pearson IAL WMA11 • January 2021 • Q8 (a) $x > 4$ (b) $9x^3 - 48x^2 + 52x - 16$ (c) $PQ = \frac{4}{3}\sqrt{3}$
- Q8** Pearson IAL WMA11 • October 2021 • Q6 (a) $(-1, 0)$, touches $(3, 0)$, $(0, 18)$ (b) $2x^3 - 10x^2 + 6x + 18$ (c) $y = \frac{512}{27}$ (gradient 0)
- Q9** Pearson IAL WMA11 • June 2025 • Q7 (a) $f(x) = \frac{1}{5}x^2(x - 4)$ (c) $(0, 0)$, $(6, 14.4)$, $(-8, -153.6)$

IDEA 02 • RECIPROCAL GRAPHS

- Q10** Dr Eslam • Asymptotes (a) $x = 2$, $y = 0$ (b) $x = 0$, $y = 4$
no intercepts; asymptotes $x = 0$, $y = 0$
- Q11** Practice • Easy asymptotes $x = 0$, $y = 0$; both branches above the x -axis
- Q12** Practice • Medium $k = 36$: $y = \frac{36}{x}$
- Q13** Practice • Medium (b) x -int $(1, 0)$, asymptote $y = k$ (c) $10 - 2\sqrt{21} < k < 10 + 2\sqrt{21}$
- Q14** Pearson IAL WMA11 • January 2021 • Q6 (b) 2 roots: the curves cross twice
- Q15** Pearson IAL WMA11 • October 2022 • Q6 (a) $x = -2$ (b) $x(x + 2)^2$ (d) 2 solutions
- Q16** Pearson IAL WMA11 • October 2023 • Q4

IDEA 03 • INTERSECTIONS

- Q17** Practice • Easy the curves cross twice (once in each of quadrants 1 and 3)
- Q18** Practice • Medium (b) 2 solutions
- Q19** Pearson IAL WMA11 • October 2019 • Q3 (a) $M\left(\frac{5}{2}, \frac{27}{4}\right)$ (b) $N\left(\frac{26}{5}, \frac{351}{25}\right)$
- Q20** Pearson IAL WMA11 • June 2021 • Q5 (a) 16.8 million (b) 53 million (c) $10 - \sqrt{58} < t < 10 + \sqrt{58}$, i.e. $2.38 < t < 15$
- Q21** Pearson IAL WMA11 • January 2022 • Q4 (a) $P(1, 5)$, $Q\left(\frac{5}{2}, \frac{7}{2}\right)$

IDEA 04 • TANGENT, CROSS OR MISS

- Q22** Dr Eslam • Tangent $x^2 - 3x + k = 0: 9 - 4k = 0 \Rightarrow k = \frac{9}{4}$
- Q23** Pearson IAL WMA11 • October 2019 • Q6 (a) $y = k$, x-int $\left(-\frac{4}{k}, 0\right)$ (b) $k = 10 \pm 4\sqrt{2}$
- Q24** Pearson IAL WMA11 • January 2022 • Q10 (a) x-int $\pm\frac{1}{3}$, asymptotes $x = 0, y = -9$ (b) $-\frac{81}{4} < k < 0$
- Q25** Pearson IAL WMA11 • January 2024 • Q7 (a) y-int $-\frac{4}{k}$, asymptote $x = k$ (b) $5 < k < 13$
- Q26** Pearson IAL WMA11 • June 2024 • Q8 (a) $(-2, 0), (0, 0), (2, 0)$ (c) $0 < A < 4$
- Q27** Pearson IAL WMA11 • October 2024 • Q6 (a) $x = 2, (0, \frac{1}{2})$ (b)(ii) $k \leq 1$ or $k \geq 4$
- Q28** Pearson IAL WMA11 • January 2025 • Q7 (a) x-int $\frac{2}{k}$, asymptote $y = -k$ (b) $0 < k < 2$ or $k > 18$
- Q29** Pearson IAL WMA11 • October 2025 • Q6 (a) asymptote $x = 2k$, y-int -2 (b) $0 < k \leq \frac{1}{2}$ or $k \geq \frac{9}{2}$
- Q30** Pearson IAL WMA11 • June 2026 • Q10 (a) y-int $-\frac{3}{2}$ (c) $k = 12: P(18, -6)$